

A Project Report On

AUTOMATED NIGHT LAMP

Submitted in partial fulfillment of the requirement for the award of Degree of

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS & COMMUNICATION ENGINEERING

SUBMITTED BY

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CERTIFICATE

This is to certify that the project entitled

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Is record of bonafide work carried out by

MEHMOOD ALI

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Under oursupervision and that no part of this project has been submitted for the award of any other Fellowship or similar titles or prizes and that the work has not been published in scientific or popular journal or magazine.

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ABSTRACT

An effort was launched during an academic engineering assignment to suggest an automatic night lamp. An automated night lamp was built using an ATMEGA328P microprocessor interfaced with an LDR and relay module to save power by automatically turning off the light during the day and turning on the light at night.

The system was designed with a buzzer to enable the system to warn when it detected dawn and turn off immediately at a defined period. It took a lot of effort to apply the information and use the finest ways. Research articles and relevant periodicals were recommended for obtaining knowledge and skills related to the present undertaking.

Background research was used to understand the use of various approaches and electronics engineering techniques. An automatic night lamp was designed with the goal of conserving electricity, and it was extensively studied about connected systems and the framework

to clearly comprehend the extent of the present work. As a result, a technique was developed to provide an automated lamp system with an integrated alert using an ATMEGA328P controller IC and an LDR sensor.

The schematic design of the system was done using an electronic circuit design tool by merging various components and resources connected with electronics engineering. The technique was structured to incorporate activities for system design and component selection. For circuit design, components were chosen systematically based on their task relevance, which aided in achieving the intended design and meeting the aim, and, therefore, the requirements of the task.

The concerns linked to the wattage of electric power were solved via this task effort, and to conserve it, an innovative controlled night lamp employing ATMEGA328P processor interfaced with LDR sensor was created. The schematic design of the system was done using an electrical circuit design tool by combining diverse components such as an LDR sensor, a relay, a 555-timer IC, a buzzer, resistors, LEDs,

capacitors, transistors, diodes, and an ATMEGA328P controller IC, among others.

A battery was also installed for power circuit backup and constant supply. The system is controlled by an ATMEGA328P controller IC, which was built with suitable pin-configuration and programming code created in the Arduino IDE using the conceptual knowledge of Embedded C language.

The Proteus simulator was used to examine the circuit performance, and the power required for the planned circuit was applied in accordance with how the linked circuit elements work. Following the development of this project prototype, which included assembling hardware work items on a breadboard.

DETAILED DESCRIPTION

The lampposts utilize a maximum of 1000W and a minimum of 250W. A detecting device that detects moving objects can be utilised to automatically adjust the lamppost's electric supply to mitigate this loss. This cuts down on the amount of time they are left on. The issue of natural resource depletion due to overconsumption is serious, and this strategy can help fix it to a considerable extent.

The approach taken here is simple to execute and cost-effective. However, this cannot be implemented on highways or other routes where traffic is usually heavy. The most significant disadvantage is that the frequent voltage fluctuations may cause the bulbs to fail. According to certain research, Vehicle headlights are extremely dangerous while driving at night.

When travelling at night, most cars utilize powerful, bright beams. The person travelling in the other direction feels irritated as a result, and there is a startling glare for a little amount of time.

The high-intensity headlight beam from the other speeding vehicle from the other direction causes this. An automated headlight dimmer with a Light Dependent Resistor sensor has been built in this research to decrease oncoming cars' headlights to prevent human eye impacts. Detecting the light automatically shifted the high beam to the low beam, lessening the glaring impact.

Essentially, this device was built on an electromagnetic program similar to an electrical bell, but it lacked a steel drum. These gadgets were commonly used as a range of sounds and then were mounted on walls and floors. Another option for using Central air devices was to develop the technology that turns the AC into a loud enough sound to power loudspeakers and then link it all to the device.

These were typically linked to "driver" circuits that adjusted the resonant frequencies or cycled them again. A Sonalert, a high-pitched clay piezoelectric warning signal, is more widely used. Changes in technology have been made to headlights, interior surfaces, and the highway environment to lessen glare directly or indirectly on the driver.

Headlights on vehicles have necessitated a balance between giving enough light for drivers to see the road ahead while avoiding excessive glare. This technique reduces human mistakes, allowing the driver to concentrate on properly traversing the road rather than manually altering headlamp settings. The principle of this work will be implemented in automobiles in the future, and it will undoubtedly be a huge success.

TECHNICAL BACKGROUND

Streetlights are often overlooked, ignored, and undervalued electrical appliances. We are not aware of their presence until everything is functioning properly. However, when it does not work, as it should, everyone notices it. When it is nighttime outside, the ideal streetlight will be on, and when it is bright, it should be off.

The municipal corporation appoints someone to carry out this role, although human mistakes are always possible. We must use automation to remove the human element from the process to address this. Because humans have become overworked and even inept, automation is envisioned as a way to reduce labour costs.

Today's home automation approaches have been introduced. *Automation* is defined as a process that follows pre-determined sequential phases with little or no human intervention. Automation is accomplished by using numerous sensors for monitoring manufacturing processes, controllers, and various methods and tools. Smart home automation provides the homeowner with complete control over his or her environment through automated lighting, dimming, and electrical items.

Because this advanced technology is utilized to automate household tasks, it is also known as home automation. An automated lamp system is one of the home automation systems that has been designed. As a result, many return to their hometowns and leave their homes for extended periods, whether they have time off work or a school holiday.

Some lights are frequently left on until the owner returns to avoid the house becoming dark. This would probably result in excessive electricity usage, which will be costly to the government. In average

cities throughout the world, lighting can account for 10–38 per cent of the entire energy bill.

Nowadays, the methods used for lighthouses, offices, and other interior areas are ineffective since a significant amount of energy is wasted during the day. This is mostly because interior lighting design assumes the worst-case scenario when the light service is provided at night, which is not usually the case. In most situations, the developed lighting system also depends on people to turn on and off the lights. One of the design requirements in Green Building is this issue.

This study presents a solution to these problems and a strategy for improving the comfort of those who utilise the interior facilities in industrial buildings. The suggested smart lighting system turns on the lights autonomously when someone is present. Lights in the proposed smart lighting system turn on automatically when someone is in the room or the inhabited space and turn off when no one is there.

In addition to this well-known method, the brightness level of the lighting may be adjusted using a desktop computer or even another smart device. In this technique, the light in the region where it needs to be monitored for better energy savings is managed automatically by the detector, and the brightness of the lamps is automatically controlled to attain the preset level that is ascertained for that room concerning the number of background lights coming from outside.

It is feasible to give higher user comfort using this strategy. It is conceivable to give improved user comfort, minimise human-forced switching on and off, and achieve successful energy savings using this technology. An Arduino controller is utilised to create the controller and display the findings. The percentage of energy savings gained by applying the suggested smart lighting controllers was calculated using economic analysis. The energy efficiency requirement for a standard-sized office was developed due to the economic study.

An automated lighting system has the obvious benefit of lowering energy usage and maintenance expenses. Because an intelligent

lighting system considers the outdoor daylight entering into the space and the occupancy state of the room, energy consumption is minimised.

Furthermore, maintenance costs are reduced since the lifespan of the light bulbs is effectively utilised, extending the series' lifespan. In addition, the suggested solution has the secondary benefit of allowing the nation to transport more oil and gas because the amount of fuel required to create power will be lowered due to the energy efficiencies generated by the intelligent lighting system.

Another benefit of implementing smart management for the indoor lighting system is reducing pollution. When less energy is utilised, the quantity of carbon dioxide generated by power-producing plants is lowered. There are numerous approaches to installing smart lighting structures to deliver lighting that is more efficient, help control the interior lighting, and minimise the operational costs of the lighting system.

The first technique is to install occupancy sensors in businesses, houses, and other places. A sensor is utilised to determine occupancy to regulate lights in this manner. If there is someone in the room, the lights will turn on; otherwise, they will turn off. This is a straightforward measure to reduce energy usage, but it is not the best solution because it ignores the contribution of background lighting entering the building.

As a result, it cannot be called a high method to regulate the indoor lighting concentration for large administration residences that operate primarily during the day.

To finish this part, it should be noted that most spaces are overlit since background light is not taken into account throughout the design process. Furthermore, lights are turned on in places where no one is present, resulting in energy waste. As a result, the Intellect Lighting System is critical in overcoming this challenge by controlling indoor lighting intensity while considering occupancy status and background light entering the room to efficiently alter the amount of illumination.

As a result, it is important to note that using appropriately selected LED bulbs in a Smart Lighting System saves energy usage and maintenance costs, pollution created by power plants, and enhances the chance of gas sales.

Adjusting the lighting system in reaction to the ambient light outdoors is perhaps the most reasonable alternative. This is the purpose of study in a lighting system wherein the street lamps are turned off if no detection systems are present when it is daylight but stay dull elsewhere. By appropriately organising the microcontroller Uno, IR obstacle detection detector, LDR, and Electrical components, the suggested design seeks to successfully substitute any hand-controlled light sources.

In this case, street lights may be electronically activated to fulfil the relevant needs when the intensity of daylight intrudes on LDR. A much more important feature is installing a monitor to maintain a record of the progress of automobiles and items that travel through the pathway. A much more important feature is installing a monitor to

maintain a count of the information of cars and items that travel through the pathway.

The quantity of automobiles passing across the roadway is recorded and displayed on the Arduino IDE's microprocessor. Moreover, high-flow street lamps are being substituted with LEDs to reduce power use even more. An automated outdoor lighting network conserves energy while reducing the chances of harm, illegal conduct, and servicing costs.

DATA TESTING

An innovative automated night lamp circuit was created and installed to save electrical power by automatically turning on or turning off the lamp when it sensed the darkness in the room or a specified location.

With this conceptual design, the system was designed so that the lamp would stay on until the light sensor (i.e. LDR) detected daylight in the morning; however, to accomplish the control mechanism, the LDR sensor was interfaced with the ATMEGA328P controller IC, and LED was used as the night lamp to provide bright light in the room whereas the significant components presented with blocks in below given functional blocks in the system design diagram-

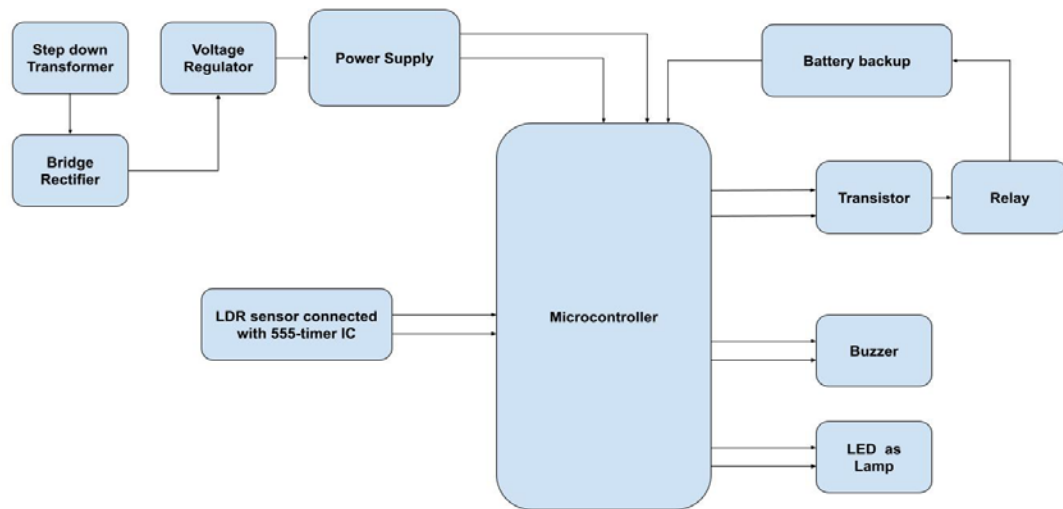


Fig. 1: System block diagram

A system-integrated alarm is activated when the sensor detects daylight in the morning. A power supply circuit was constructed to provide power to the system by incorporating a step-down transformer, a bridge-type full-wave rectifier with diodes for rectification, and a regulator IC to provide regulated DC to the controller circuit.

A battery was also included in the circuit for backup and continuous power supply in the event of a mains breakdown. Depending on the availability of the main supply, the battery would be charged using the given diode and resistor. In the event of a power outage, the

battery would automatically take over the load without delay, with a diode to prevent the battery from being discharged.

The schematic design of the system was accomplished using an electronic circuit design tool, and then simulation was performed to evaluate circuit functioning. Following that, a program prototype was created by assembling hardware project components on a breadboard.

The schematic design of the system was done using an electrical circuit design tool by combining diverse features such as an LDR sensor, a relay, a 555-timer IC, a buzzer, resistors, LEDs, capacitors, transistors, diodes, and an ATMEGA328P controller IC included with its standard files into the circuit design tool.

It was considered the LDR's operational characteristics since it detects the light intensity corresponding to which the resistance varies and this fluctuation in resistance, i.e. the LDR's resistance increases during darkness and lowers when full illumination is detected.

So, the LDRs were used to detect light intensity, and the LDR was linked to the controller IC through a timer (555-IC) configured in monostable mode. Further to define the programming approach for the microcontroller an algorithm defined that discussed here in figure 2 in terms of procedure flow diagram-

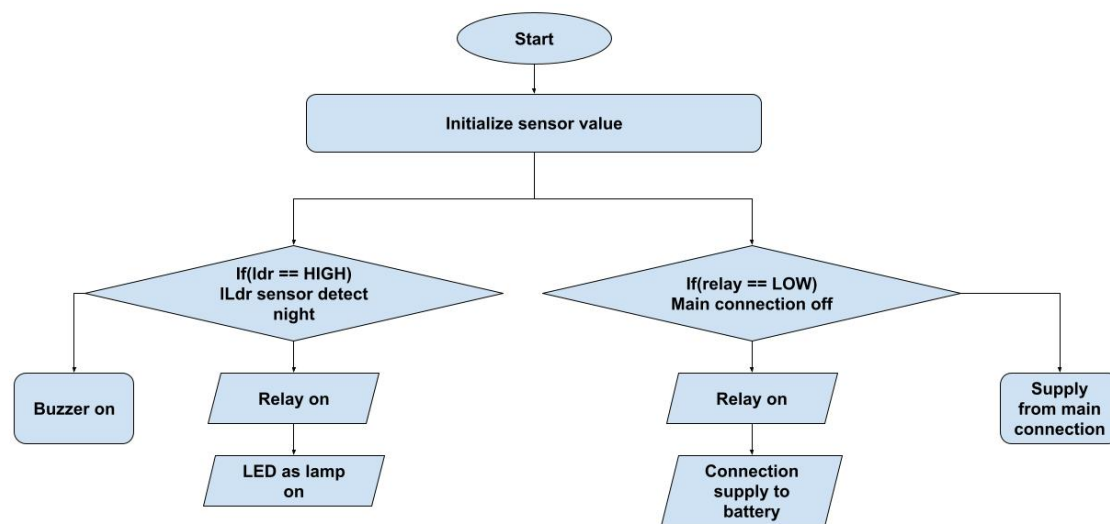


Fig. 2: Procedure flow diagram

By referring to the figure 2 the required program developed using the concepts of embedded C language, however the needed header files mentioned at the beginning of the program. The Arduino IDE workspace along with compilation shown in following figures-

```

int buzz_fun = 2;
int night_lmp = 3;
int lmp_led = 13;
int lmp_val;
void setup()
{
    Serial.begin(9600);
    pinMode(buzz_fun, OUTPUT);
    pinMode(night_lmp, INPUT);
    Serial.print("Automated night lamp");
    digitalWrite(buzz_fun, LOW);
    digitalWrite(lmp_led, HIGH);
}

void loop()
{
    lmp_val = digitalRead(night_lmp);
    Serial.print(lmp_val);
    delay(100);
    if (lmp_val == HIGH)
    {
        digitalWrite(buzz_fun, HIGH);
        digitalWrite(lmp_led, LOW);
        Serial.print("Ldr detect night");
    }
}

```

Done compiling.

Fig. 3: Arduino IDE workspace

```

void loop()
{
    lmp_val = digitalRead(night_lmp);
    Serial.print(lmp_val);
    delay(100);
    if (lmp_val == HIGH)
    {
        digitalWrite(buzz_fun, HIGH);
        digitalWrite(lmp_led, LOW);
        Serial.print("Ldr detect night");
    }

    if (lmp_val == LOW)
    {
        digitalWrite(buzz_fun, LOW);
        digitalWrite(lmp_led, HIGH);
        Serial.print("Ldr detect day");
    }
}

```

Done compiling.

Fig. 4: Arduino IDE workspace

Thus the system is controlled by an ATMEGA328P controller IC, which was built with suitable pin-configuration and programming code created in the Arduino IDE using the embedded C language. According to the design methodology the schematic design and implementation conducted with respect to the pin-specifications whereas in the following figures the schematic and simulation performed in the Proteus simulator-

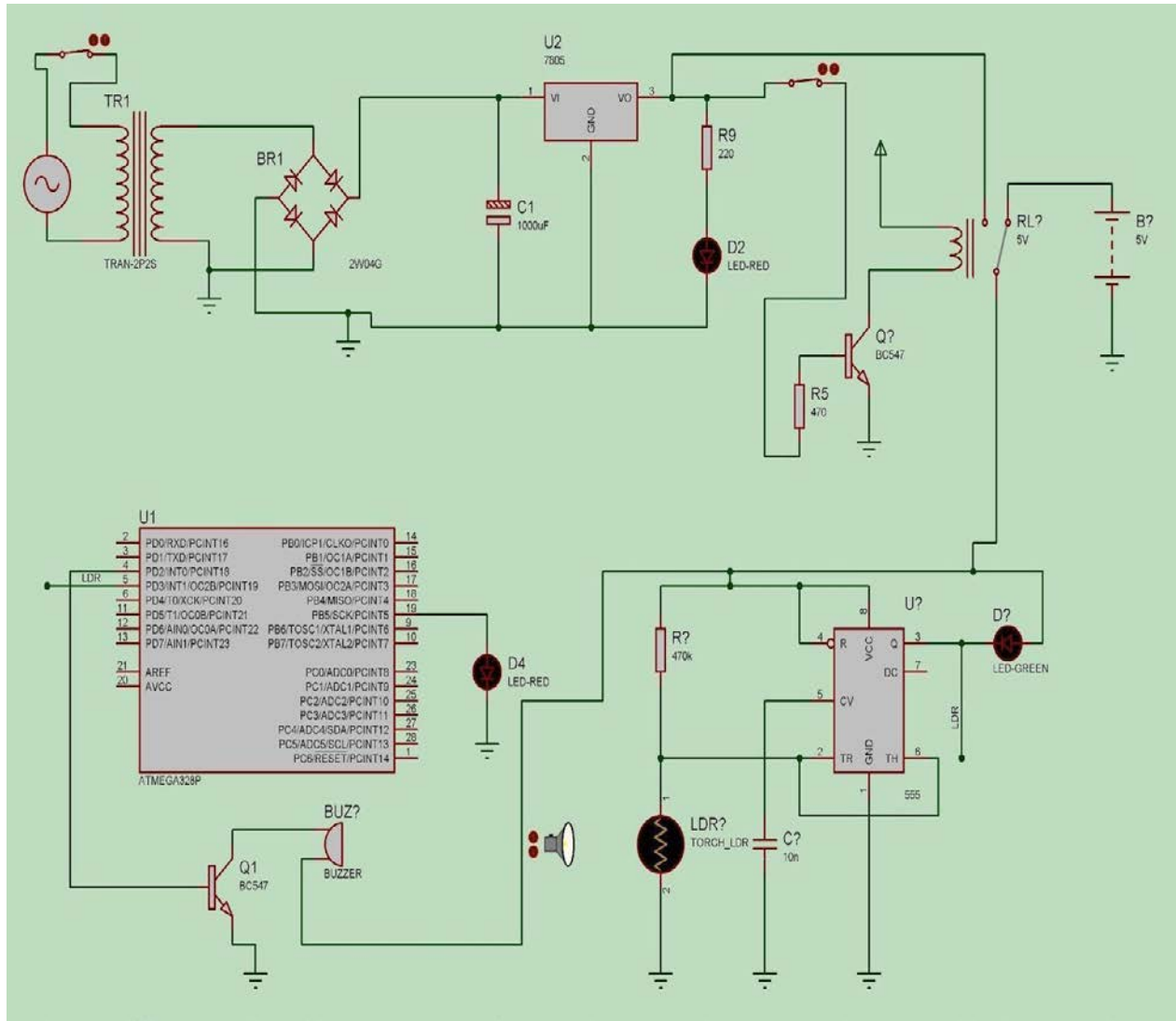


Fig. 5: Schematic design and simulation analysis

In context to the schematic design & simulation analysis the needed power is specified in the preceding circuit simulation diagrams, and the system is suitably initiated; however, when the first LDR work is lighted with the room's ambient light, its resistance reduces, and the timer IC is triggered at positive potential. Furthermore, low power at

the IC results in the LED remaining off. On the other hand, the device resistance increases when the initial LDR illumination is reduced. For continuous operations, a low-value capacitor was used to preserve a monostable state of the timer IC; the timer effect of the IC was abolished as a result of this circuit setup.

The buzzer was specifically attached to the controller IC via a transistor and served as an alarm in the intended system. Therefore as per the schematic design and simulation analysis it is depicted that red LED on for indicating the supply of power as shown here-

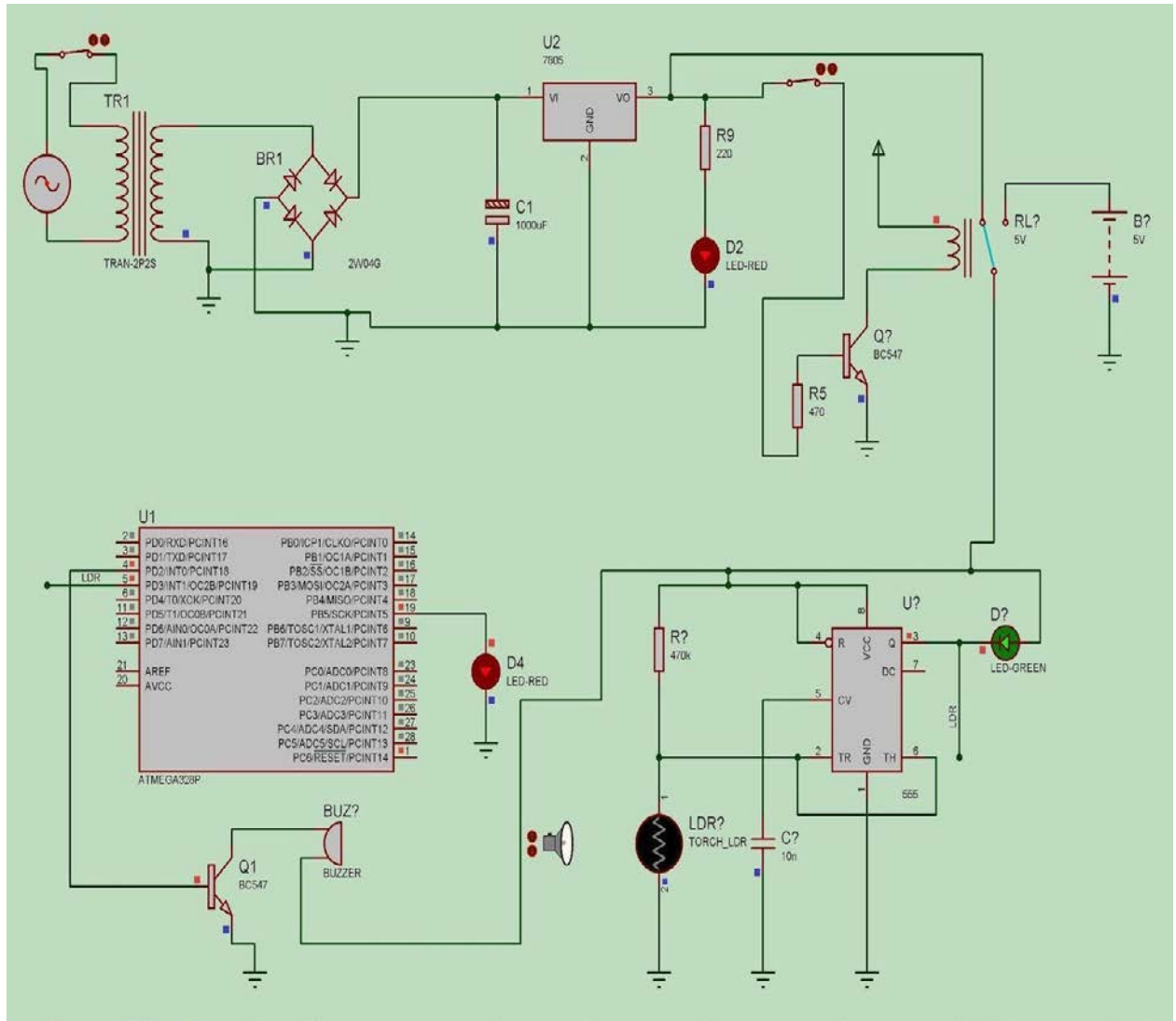


Fig. 6: Schematic design and simulation analysis

The pin numbers were established to connect the buzzer, LED lamp, and light sensor with the microcontroller using the programming of the microcontroller (ATMEGA328P) IC, and the optimum system design was obtained after working attentively and utilizing the right

design method.

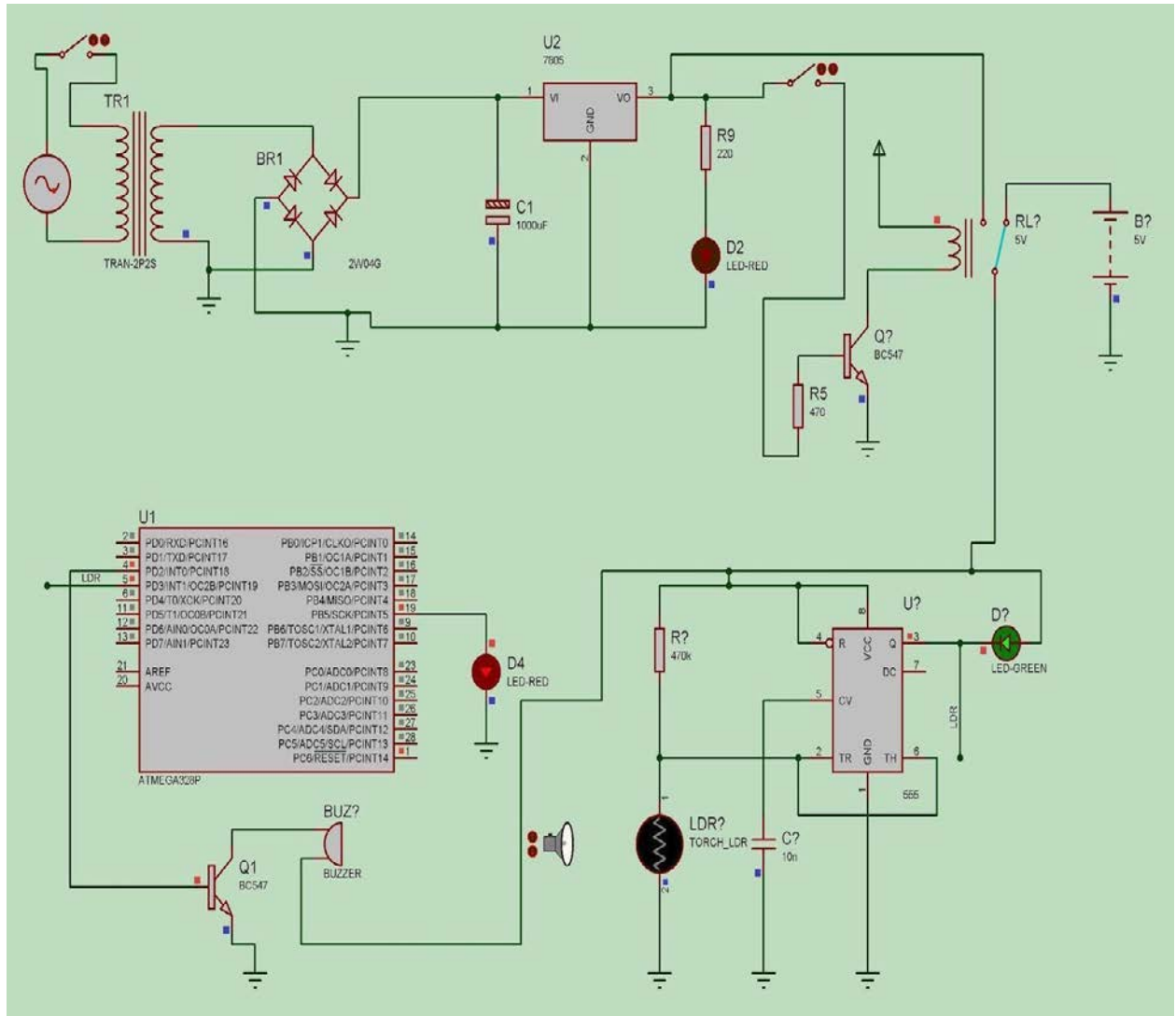


Fig. 7: Schematic design and simulation analysis

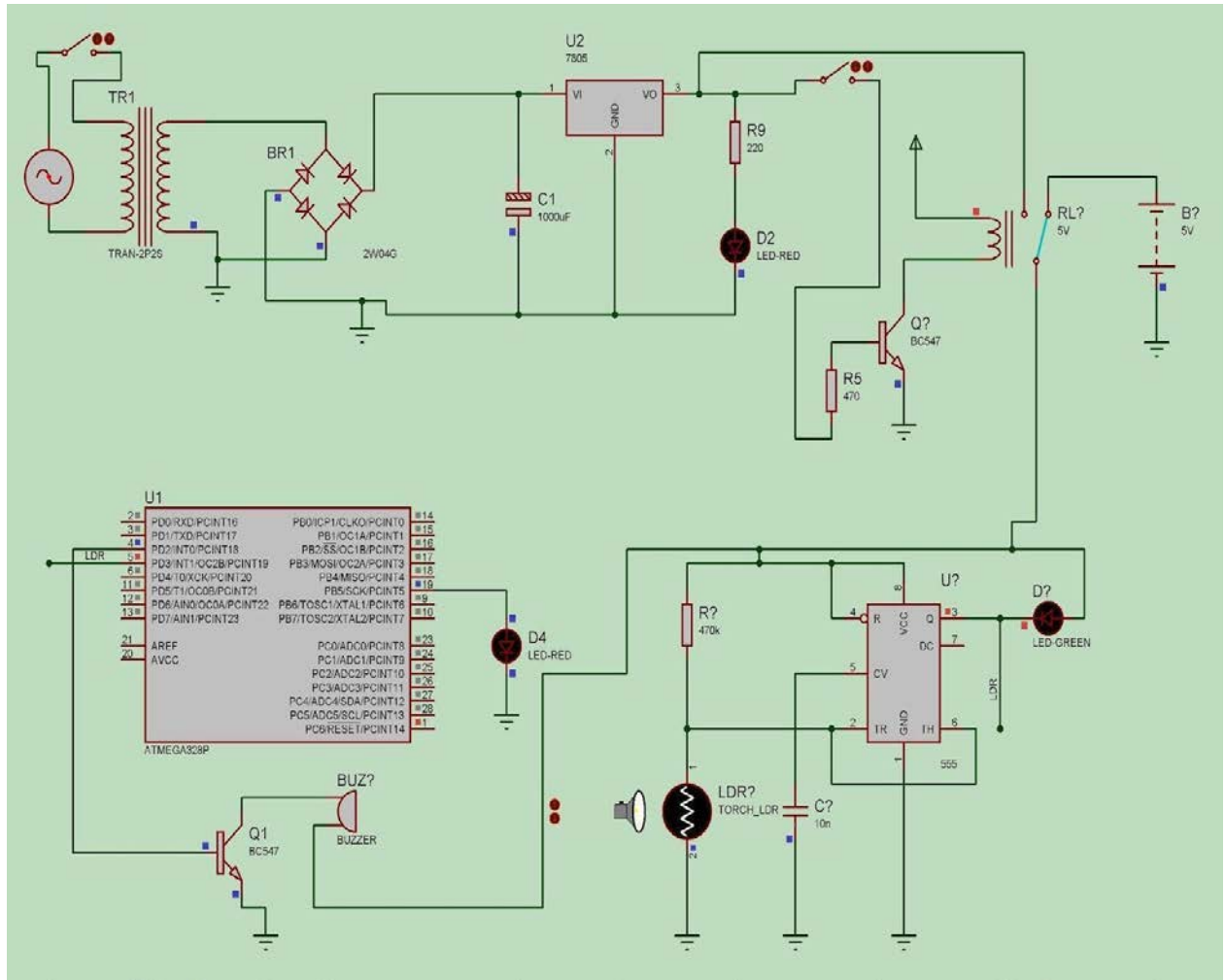


Fig. 8: Schematic design and simulation analysis

The circuit performance was evaluated using the Proteus simulator, and the power required for the schematic diagram was implemented in accordance with how the attached circuit components operate. To demonstrate the working of the prototype model, all elements were assembled on a breadboard for temporary circuit connections, and several jumper wires were used to achieve it.

After receiving electricity, the prototype model was set up to detect light intensity for day or night, and if the day was detected, the system would turn off the night lamp and turn on the alarm, and if the night was detected, the system would turn on the room lights. In below figures the prototype modeling illustrated as-

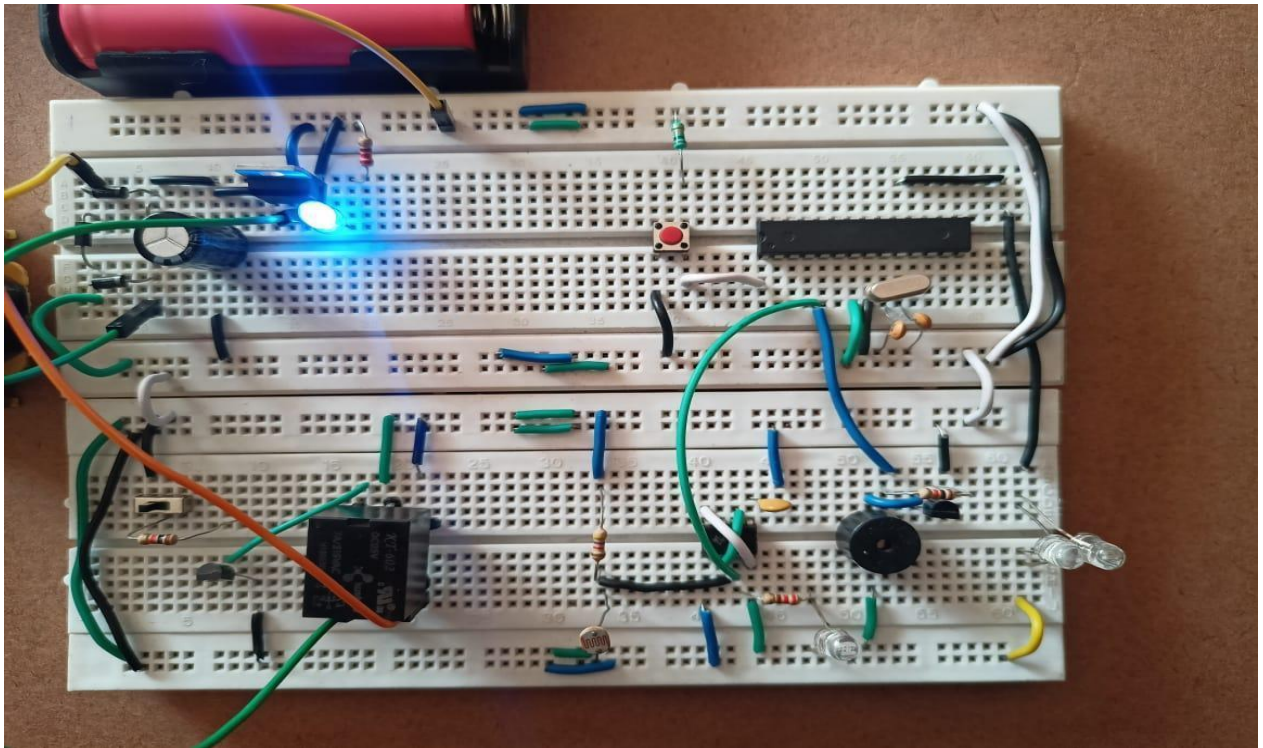


Fig. 9 : Prototype modelling

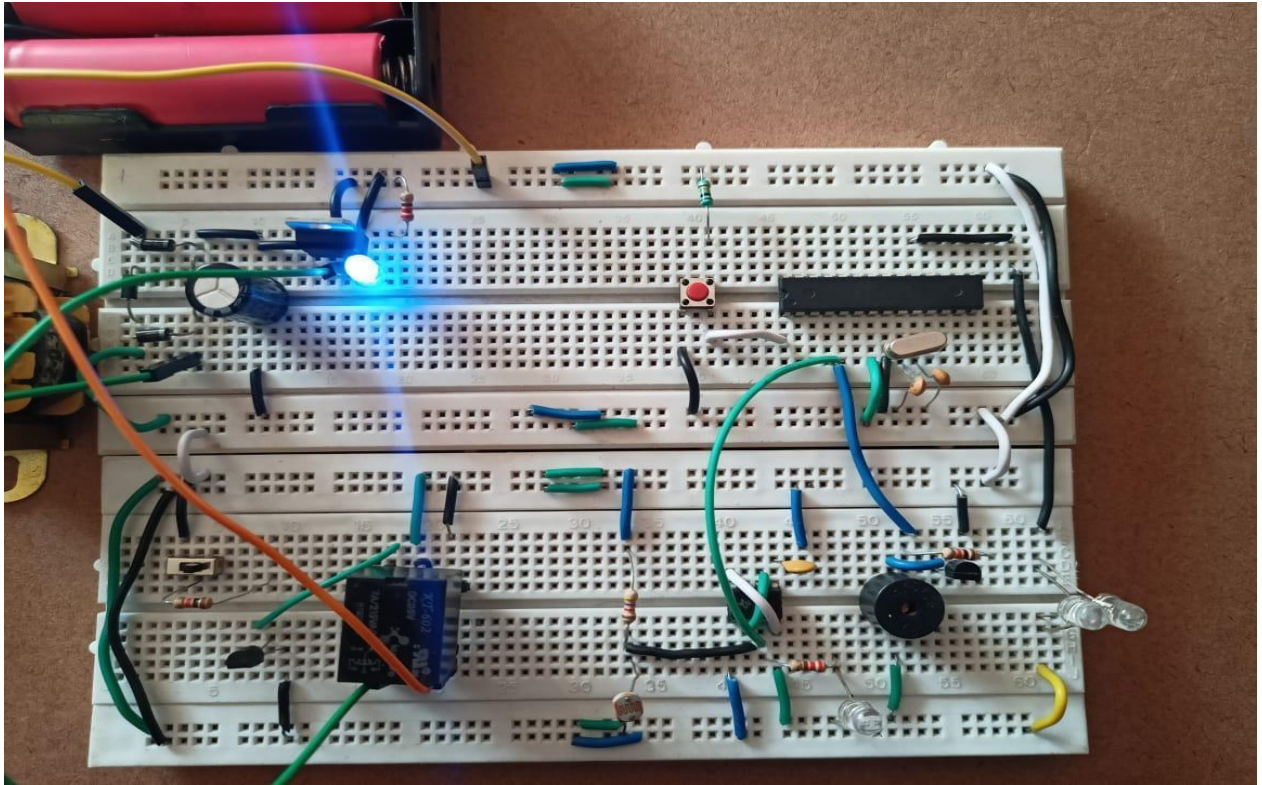


Fig. 10 : Prototype modelling

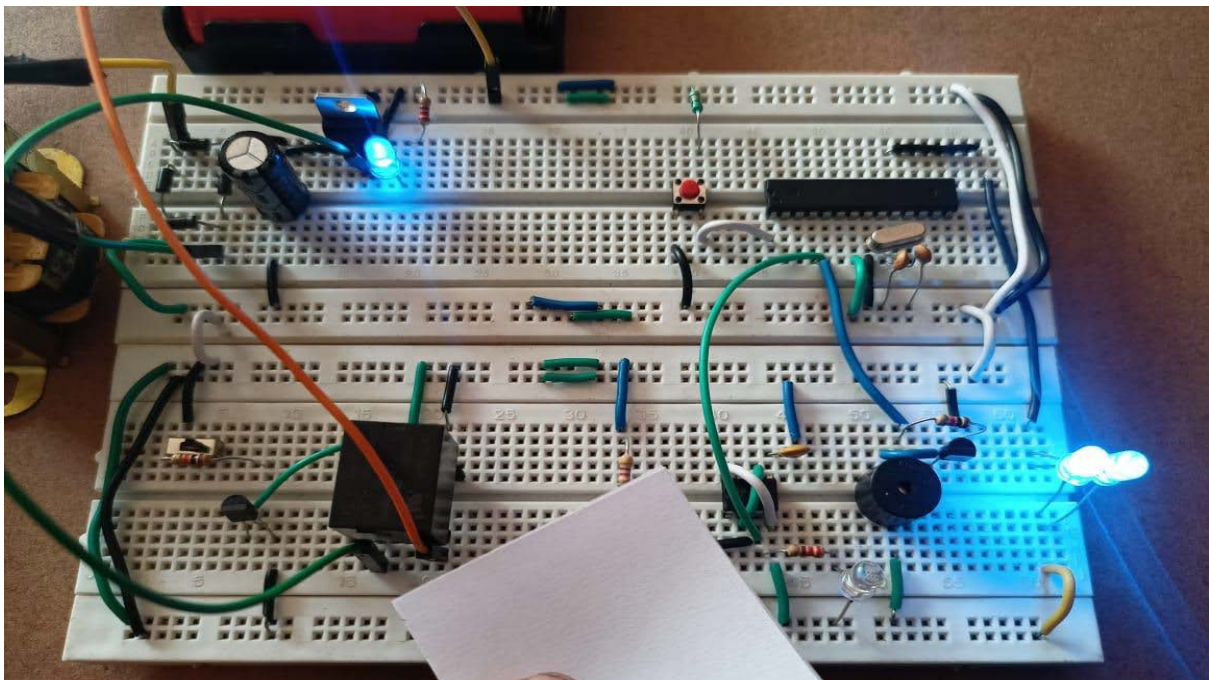


Fig. 11 : Prototype modelling

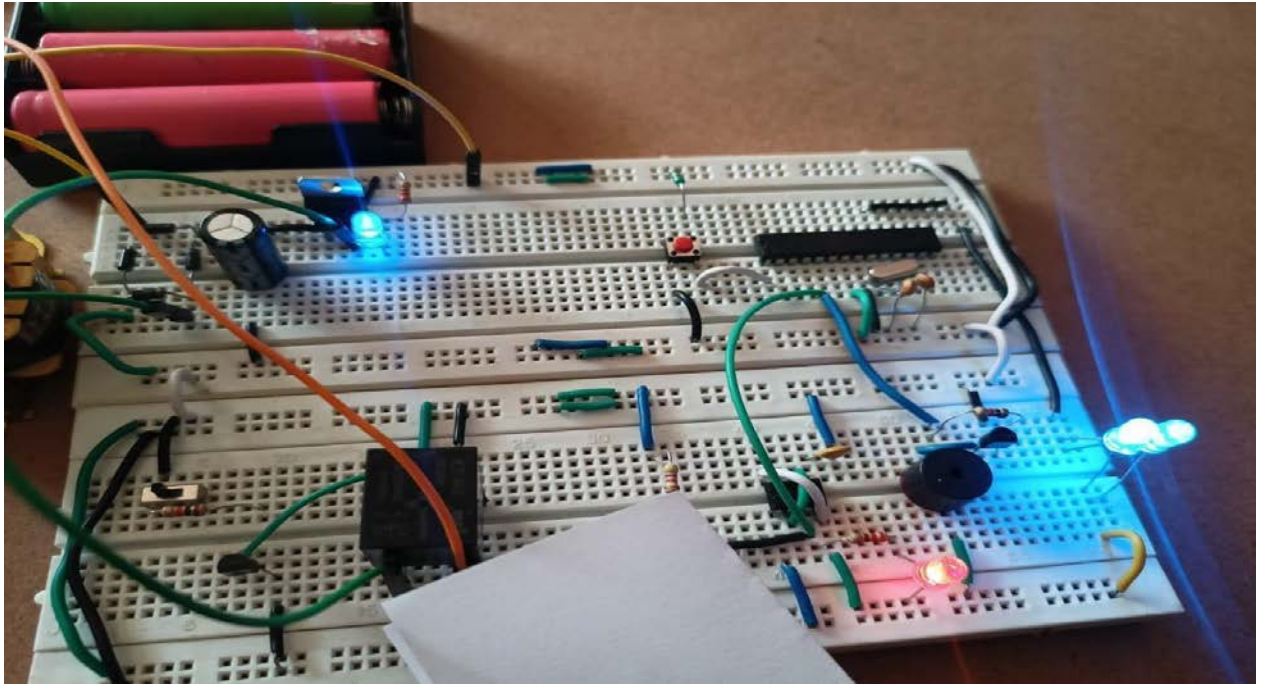


Fig. 12 : Prototype modelling

For analysing the developed prototype model a paper is considered to cover the sensor as illustrated in figure 11 & figure 12 due to which the intensity of light varies and respectively the connected lamp (LED) tuned on or off. Thus the prototype modeling was completed and demonstrated successfully.

The best available methodologies and technology were used to tackle the complexities encountered during the task. However, typical design methodologies and engineering foundations were implemented as per

the established plan, a prototype of the LDR was successfully built, and microcontroller configured an automated type night lamp for the room.

Because constructing a light sensor circuit for this project was challenging, the timer IC (555 IC) was used, and fluctuations in the resistance value of the LDR were translated into voltage and delivered to the IC. The LDR sensor and the other electronic components in the system were also appropriately linked to create an ideal system. The complete system architecture was described with precision, with no inaccuracies.

ANALYSIS & CONCLUSION

The knowledge and abilities were applied to get exceptional outcomes and effectively finish the project job. To continue the work quickly and comprehend the principles of the engineering solution, the conceptual understanding was combined with theoretical aspects. Strategic plans were created using strategy and tactics, and work was carried out in a structured manner.

Deep analysis was done to improve technical knowledge and skills to cope with technical issues that arose throughout the task. The most up-to-date methods were employed to efficiently complete the duties according to the set scope. With engineering expertise, many options for overcoming the hurdles were investigated. Several techniques

were successfully used with the practical knowledge and skills during the task.

The project's goal was met exactly as planned, and the accountabilities were handled flawlessly. It was performed under the current work and provided the system engineering whose key component was a microcontroller (ATMEGA328P) as the total night lamp control was dependent on the controller.

The automated system was created so that energy waste may be minimized to the greatest extent possible. A model of the LDR was created, and a microprocessor was used to program an autonomous type night lamp system for the room, which then executed the required operations.

The components used in the design process were fully described, together with their frameworks and application logic. Significant engineering information was gathered, which would aid in the completion of comparable tasks. The task was successfully done.

BIBLIOGRAPHY APPLIED

1. Applied Mechanics and Mechatronics Automation, 2012
2. Automation and Systems Issues in Air Traffic Control, 2012
3. Manufacturing Engineering and Automation II, 2012
4. Stagecraft Fundamentals A Guide and Reference for Theatrical Production Rita Kogler Carver , · 2012
5. Electrical Energy Efficiency Technologies and Applications Andreas Sumper, Angelo Bagгинi , · 2012